Making legal expertise repeatable and traceable

My understanding of the Knowledge Systems & AI Automation internship

BUSINESS CONTEXT

Documents people depend on

Docitalis describes legal translations for Italian citizenship, with document preparation, review and certification.

ROLE CONTEXT

Lawyer-approved knowledge

The brief places discovery with legal teams at Aprigliano: extract facts, arguments, conditions and exceptions.

ENGINEERING NEED

Explicit rules, clear decisions

Turn approved reasoning into explicit rules, versioned modules and reviewable software changes.

My focus: make expert decisions explicit, preserve their sources, and test every change.

Sources: Docitalis website; supplied role description. Proposed approach, not an internal-system assessment.

Rules assemble. Retrieval supports review.

Legal strategy remains explicit and lawyer-approved; a language model does not select the argument.

APPROVED INPUTS

Case facts + strategy
Knowledge version

→

DETERMINISTIC ENGINE

Rules + dependencies
Ordered modules

→

REVIEWABLE OUTPUT

Draft + decision trace
Version + output hash

SOURCE MATRIX + ASSISTIVE REVIEW

Retrieve provisions · flag missing citations
Identify modules affected by recorded judgments

→

LAWYER ASSESSMENT

Review findings
Assess applicability

Source: supplied role description. Diagram is a proposed conceptual architecture.

DAY-TO-DAY CONTRIBUTION

From one approved example to one tested change

Start with a narrow workflow and resolve ambiguity with the legal team.

01	DISCOVER	**Walk through an approved pleading**	Capture facts, choices, exceptions and open questions with a lawyer.
02	FORMALISE	**Make the rules inspectable**	Define module conditions and links in a structured schema.
03	IMPLEMENT	**Change one bounded behavior**	Update Python logic and preserve versions and decision traces.
04	VALIDATE	**Review evidence before release**	Compare expected outputs, explain differences and record approval.

Python + structured data + Git; use formal constraint checks where they add value.

Source: supplied role description. Activities and deliverables below are proposed.

VALIDATION · IMPACT ANALYSIS

A source update should trigger a reviewable change

Example: a new judgment is linked to a provision already used by document modules.

Record source

Authority, date, jurisdiction, version

→

Find dependencies

Which modules cite or rely on it?

→

Run regression

Compare against approved examples

→

Review differences

Lawyer confirms what should change

What the checks protect

Reproducibility

Expected module order and citations

Dependencies, exclusions and update alerts

APPEALS EXTENSION

Map findings, reasoning and authorities for a lawyer to assess potential appeal grounds.

Source: supplied role description; illustrative update, not a real judgment or legal conclusion.

EVIDENCE · FIRST CONTRIBUTION

Small, explainable work before broader automation

My earlier research contribution and recent demonstrations are distinct.

WHAT I CAN DISCUSS

Earlier research: literature review, problem identification and initial design.

UkilRAG: rules, traces and synthetic regressions.
LearnRAG: real model outputs and limitations.

View UkilRAG ↗ View LearnRAG ↗

PROPOSED FIRST MONTH

Understand one workflow with the team.
Agree on examples and expected behavior.
Model and implement one bounded rule set.
Review test results and a baseline measure.

Measure review time, rework and unexpected output changes before setting improvement targets.

Evidence: github.com/nafismam/UkilRAG and github.com/nafismam/LearnRAG. No production or legal-accuracy claim.